INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 02: Technical Ability (Quantitative Aptitude) Compendium (70 Questions)

Section: 02_Mathematical_Ability | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 02: Technical Ability (Quantitative Aptitude) Compendium (70 Questions)

Infosys System Engineer ITM Mock Test Series (Papers 01 to 07)

> **Curriculum Overview:** Section 2 tests technical quantitative aptitude, including permutations and combinations, probability, work and time, pipes and cisterns, speed/trains/races, commercial mathematics, mixtures and alligations, and interest formulas.
> **Total Questions:** 70 Questions (10 Questions per Paper × 7 Papers)
> **Target Time:** 35 Minutes per 10-question section (3.5 min/question)
> **Verification Status:** 100000000% Verified Solutions & Zero Hallucinations.

Question ITM-P01-Q16 (Infosys System Engineer ITM Mock Paper 1 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?

Options:
- **(A)** 5400
- **(B)** 3600
- **(C)** 10800
- **(D)** 2700

Correct Answer: Option **(A)** — `5400`

Detailed Step-by-Step Solution & Mathematical Derivation:
Word COMMITTEE contains 9 letters: C(1), O(1), M(2), I(1), T(2), E(2).
Vowels: O, I, E, E (4 vowels, with two E's).
Consonants: C, M, M, T, T (5 consonants, with two M's and two T's).
1) Arrange the 5 consonants first:
Number of ways = 5! / (2! × 2!) = 120 / 4 = 30 ways.
2) 5 consonants create 6 available gaps: _ C _ M _ M _ T _ T _
To ensure no two vowels are adjacent, choose 4 gaps out of 6 for the 4 vowels:

C(6, 4) = 15 ways.
3) Arrange the 4 vowels (O, I, E, E) in the chosen 4 gaps:
Number of ways = 4! / 2! = 24 / 2 = 12 ways.
Total arrangements = 30 × 15 × 12 = 5,400.

Question ITM-P01-Q17 (Infosys System Engineer ITM Mock Paper 1 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?

Options:
- **(A)** 90
- **(B)** 100
- **(C)** 105
- **(D)** 120

Correct Answer: Option **(B)** — `100`

Detailed Step-by-Step Solution & Mathematical Derivation:
Committee of 5 from 5 boys and 4 girls with at least 2 boys and at least 2 girls.
Possible compositions (Boys, Girls):
- Case 1: 2 Boys and 3 Girls => C(5, 2) × C(4, 3) = 10 × 4 = 40 ways.
- Case 2: 3 Boys and 2 Girls => C(5, 3) × C(4, 2) = 10 × 6 = 60 ways.
Total possible ways = 40 + 60 = 100 ways.

Question ITM-P01-Q18 (Infosys System Engineer ITM Mock Paper 1 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?

Options:
- **(A)** 4/11
- **(B)** 1/3
- **(C)** 5/11
- **(D)** 4/9

Correct Answer: Option **(A)** — `4/11`

Detailed Step-by-Step Solution & Mathematical Derivation:
Two fair dice thrown. Total outcomes with at least one 4:
Sample space S = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (5,4), (6,4)} => n(S) = 11 outcomes.
Sums for each outcome:

- (4,1) -> 5 (prime)
- (4,2) -> 6
- (4,3) -> 7 (prime)
- (4,4) -> 8
- (4,5) -> 9
- (4,6) -> 10
- (1,4) -> 5 (prime)
- (2,4) -> 6
- (3,4) -> 7 (prime)
- (5,4) -> 9
- (6,4) -> 10

Favorable outcomes where sum is prime: {(4,1), (4,3), (1,4), (3,4)} => 4 outcomes.
Probability = 4 / 11.

Question ITM-P01-Q19 (Infosys System Engineer ITM Mock Paper 1 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?

Options:
- **(A)** 3/11
- **(B)** 1/4
- **(C)** 2/11
- **(D)** 5/22

Correct Answer: Option **(A)** — `3/11`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total balls = 5 red + 4 blue + 3 green = 12 balls.
3 balls drawn at random without replacement.
Total ways = C(12, 3) = (12 × 11 × 10) / (3 × 2 × 1) = 220.
Favorable ways to get one of each colour = C(5, 1) × C(4, 1) × C(3, 1) = 5 × 4 × 3 = 60.
Probability = 60 / 220 = 3 / 11.

Question ITM-P01-Q20 (Infosys System Engineer ITM Mock Paper 1 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:

Options:
- **(A)** 7.1%
- **(B)** 5.5%
- **(C)** 8.6%
- **(D)** 6.4%

Correct Answer: Option **(A)** — `7.1%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let Cost Price (CP) = 100.
Marked Price (MP) = 100 × 1.40 = 140.
Successive discounts of 10% and 15%:
Selling Price (SP) = 140 × (1 - 0.10) × (1 - 0.15) = 140 × 0.90 × 0.85 = 126 × 0.85 = 107.1.
Net profit = 107.1 - 100 = 7.1%.

Question ITM-P01-Q21 (Infosys System Engineer ITM Mock Paper 1 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, … (one person per day, starting with A). On which day is the job completed?

Options:
- **(A)** 16th
- **(B)** 17th
- **(C)** 18th
- **(D)** 19th

Correct Answer: Option **(B)** — `17th`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let total work = LCM(12, 18, 24) = 72 units.
Daily efficiencies:
- A = 72 / 12 = 6 units/day
- B = 72 / 18 = 4 units/day
- C = 72 / 24 = 3 units/day

They work on alternate days: A on Day 1, B on Day 2, C on Day 3, A on Day 4, etc.
In 1 cycle of 3 days, work done = 6 + 4 + 3 = 13 units.
In 5 full cycles (15 days), work done = 5 × 13 = 65 units.
Remaining work = 72 - 65 = 7 units.
- Day 16: A works and completes 6 units. Total work = 65 + 6 = 71 units. Remaining = 1 unit.
- Day 17: B works. Since B does 4 units/day, B finishes the remaining 1 unit in 1/4 of a day.

Therefore, the job is completed on the 17th day.

Question ITM-P01-Q22 (Infosys System Engineer ITM Mock Paper 1 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:

Options:
- **(A)** 90 km/h
- **(B)** 108 km/h
- **(C)** 120 km/h
- **(D)** 72 km/h

Correct Answer: Option **(B)** — `108 km/h`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total distance to clear = 240 m + 160 m = 400 m.
Let speed of faster train = v1 m/s and slower train = v2 m/s.
1) Running in opposite directions, relative speed = v1 + v2 = 400 / 8 = 50 m/s.
2) Running in same direction, relative speed = v1 - v2 = 400 / 40 = 10 m/s.
Adding the two equations: 2·v1 = 60 => v1 = 30 m/s.
Converting to km/h: 30 × (18 / 5) = 108 km/h.

Question ITM-P01-Q23 (Infosys System Engineer ITM Mock Paper 1 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?

Options:
- **(A)** 1520
- **(B)** 1600
- **(C)** 1360
- **(D)** 1440

Correct Answer: Option **(A)** — `1520`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total registered voters = 20,000.
Voters who voted = 80% of 20,000 = 16,000.
Invalid votes = 5% of 16,000 = 800.
Valid votes = 16,000 - 800 = 15,200.
Winner obtained 55% of valid votes => Loser obtained (100% - 55%) = 45%.

Winning margin = 55% - 45% = 10% of valid votes.
Winning margin = 0.10 × 15,200 = 1,520 votes.

Question ITM-P01-Q24 (Infosys System Engineer ITM Mock Paper 1 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.

Options:
- **(A)** ₹8,000
- **(B)** ₹9,000
- **(C)** ₹10,000
- **(D)** ₹12,000

Correct Answer: Option **(C)** — `₹10,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Difference between CI and SI for 3 years at rate r%:
Formula: Difference = $P \times (r / 100)^2 \times (3 + r / 100)$.
Given r = 10% and Difference = ₹310:
$310 = P \times (0.10)^2 \times (3 + 0.10)$
$310 = P \times 0.01 \times 3.10 = P \times 0.031$
$P = 310 / 0.031 = ₹10,000$.

Question ITM-P01-Q25 (Infosys System Engineer ITM Mock Paper 1 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?

Options:
- **(A)** ₹4,000
- **(B)** ₹5,000
- **(C)** ₹6,000
- **(D)** ₹7,000

Correct Answer: Option **(B)** — `₹5,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let sum invested at 8% SI = ₹x.
Then sum invested at 10% CI = ₹(12,000 - x).
Time = 2 years.
- Simple Interest at 8% for 2 years = (x × 8 × 2) / 100 = 0.16x.

- Compound Interest at 10% for 2 years = (12,000 - x) × [(1 + 0.10)^2 - 1] = (12,000 - x) × 0.21 = 2,520 - 0.21x.
Total interest = 0.16x + 2,520 - 0.21x = 2,520 - 0.05x.
Given total interest = ₹2,270:
2,520 - 0.05x = 2,270
0.05x = 250
x = 250 / 0.05 = ₹5,000.

Question ITM-P02-Q16 (Infosys System Engineer ITM Mock Paper 2 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?

Options:
- **(A)** 7560
- **(B)** 15120
- **(C)** 3780
- **(D)** 30240

Correct Answer: Option **(A)** — `7560`

Detailed Step-by-Step Solution & Mathematical Derivation:
MANAGEMENT has M2, A2, N2, G1, E2, T1 (10 letters). Vowels A,A,E,E form one block: internal arrangements = 4!/(2!2!) = 6. Treating the block as one unit alongside the 6 consonants (M2,N2,G1,T1) gives 7 units: 7!/(2!2!) = 1260. Total = 1260 × 6 = 7,560.

Question ITM-P02-Q17 (Infosys System Engineer ITM Mock Paper 2 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?

Options:
- **(A)** 265
- **(B)** 295
- **(C)** 310
- **(D)** 325

Correct Answer: Option **(A)** — `265`

Detailed Step-by-Step Solution & Mathematical Derivation:
0 women: C(6,4)=15. 1 woman: C(6,3)×C(5,1)=100. 2 women: C(6,2)×C(5,2)=150. Total = 15+100+150 = 265.

Question ITM-P02-Q18 (Infosys System Engineer ITM Mock Paper 2 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?

Options:
- **(A)** 3/7
- **(B)** 1/2
- **(C)** 3/8
- **(D)** 2/7

Correct Answer: Option **(A)** — `3/7`

Detailed Step-by-Step Solution & Mathematical Derivation:
Excluding TTT leaves 7 equally likely outcomes. Exactly two heads: HHT, HTH, THH = 3 outcomes. Probability = 3/7.

Question ITM-P02-Q19 (Infosys System Engineer ITM Mock Paper 2 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?

Options:
- **(A)** 99/190
- **(B)** 91/190
- **(C)** 33/95
- **(D)** 29/95

Correct Answer: Option **(A)** — `99/190`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(no defective) = (14×13)/(20×19) = 91/190. P(at least one defective) = 1 − 91/190 = 99/190.

Question ITM-P02-Q20 (Infosys System Engineer ITM Mock Paper 2 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.

Options:
- **(A)** 20%
- **(B)** 22.2%
- **(C)** 25%
- **(D)** 18%

Correct Answer: Option **(B)** — `22.2%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Take CP = 100 per unit weight, so marked price = 125 (25% intended profit). After a 12% discount, SP = 125 × 0.88 = 110. But he delivers only 0.9 units (10% short weight) whose true cost is 90. Profit = 110 − 90 =
20 on a cost of 90, giving 20/90 × 100 ≈ 22.2%.

Question ITM-P02-Q21 (Infosys System Engineer ITM Mock Paper 2 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?

Options:
- **(A)** 20 days
- **(B)** 24 days
- **(C)** 25 days
- **(D)** 30 days

Correct Answer: Option **(D)** — `30 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
In 5 days together, they finish 5/15 = 1/3. The remaining 2/3 is done by B alone in 20 days, so B's rate = (2/3)/20 = 1/30/day (B alone takes 30 days). Combined rate = 1/15, so A's rate = 1/15 − 1/30 = 1/30/day — A alone also takes 30 days.

Question ITM-P02-Q22 (Infosys System Engineer ITM Mock Paper 2 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.

Options:
- **(A)** 4 km/h
- **(B)** 5 km/h

- **(C)** 6 km/h
- **(D)** 3 km/h

Correct Answer: Option **(B)** — `5 km/h`

Detailed Step-by-Step Solution & Mathematical Derivation:
60/(15+s) + 60/(15−s) = 9. Simplifying: 1800 = 9(225 − s^2), so s^2 = 25, s = 5 km/h.

Question ITM-P02-Q23 (Infosys System Engineer ITM Mock Paper 2 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?

Options:
- **(A)** 27,000
- **(B)** 26,500
- **(C)** 26,000
- **(D)** 27,500

Correct Answer: Option **(A)** — `27,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
P × 1.10 × 0.92 = 27,324, so P × 1.012 = 27,324, giving P = 27,000.

Question ITM-P02-Q24 (Infosys System Engineer ITM Mock Paper 2 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A sum of money amounts to ₹13,000 after 3 years and to ₹15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.

Options:
- **(A)** 8%
- **(B)** 10%
- **(C)** 12%
- **(D)** 9%

Correct Answer: Option **(B)** — `10%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Interest for 2 years (year 5 − year 3) = 15,000 − 13,000 = 2,000, so 1 year's interest = 1,000. Principal = 13,000 − 3×1,000 = 10,000. Rate = 1,000/10,000 × 100 = 10%.

Question ITM-P02-Q25 (Infosys System Engineer ITM Mock Paper 2 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
₹16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ₹18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).

Options:
- **(A)** 9.5%
- **(B)** 10.5%
- **(C)** 11.5%
- **(D)** 12.5%

Correct Answer: Option **(B)** — `10.5%`

Detailed Step-by-Step Solution & Mathematical Derivation:
SI on ₹18,000 at 10% for 2 years = ₹3,600. Setting CI on ₹16,000 equal: $16{,}000\times((1+r)^2 - 1) = 3{,}600$ gives $(1+r)^2 = 1.225$, so $r \approx 10.5\%$.

Question ITM-P03-Q16 (Infosys System Engineer ITM Mock Paper 3 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?

Options:
- **(A)** 1260
- **(B)** 2520
- **(C)** 630
- **(D)** 420

Correct Answer: Option **(A)** — `1260`

Detailed Step-by-Step Solution & Mathematical Derivation:
Choose which of the 3 students gets 4 books: 3 ways. Choose that student's 4 books: $C(8,4)=70$. Split the remaining 4 books between the other two specific students, 2 each: $C(4,2)=6$. Total = $3\times70\times6 = 1{,}260$.

Question ITM-P03-Q17 (Infosys System Engineer ITM Mock Paper 3 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?

Options:
- **(A)** 658
- **(B)** 686
- **(C)** 700
- **(D)** 623

Correct Answer: Option **(A)** — `658`

Detailed Step-by-Step Solution & Mathematical Derivation:
Valid (men,women) splits with men>women, total=6: (6,0): C(7,6)×C(6,0)=7. (5,1): C(7,5)×C(6,1)=21×6=126. (4,2): C(7,4)×C(6,2)=35×15=525. Total = 7+126+525 = 658. (Cross-check: adding the equal split (3,3)=700 and women>men splits=358 gives 658+700+358=1,716=C(13,6), confirming the count.)

Question ITM-P03-Q18 (Infosys System Engineer ITM Mock Paper 3 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?

Options:
- **(A)** 4/11
- **(B)** 7/11
- **(C)** 2/5
- **(D)** 1/2

Correct Answer: Option **(A)** — `4/11`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(Red|A)=4/10, P(Red|B)=7/10, each bag chosen with probability 1/2. P(Red) = ½×0.4 + ½×0.7 = 0.55. By Bayes' theorem, P(A|Red) = (½×0.4)/0.55 = 0.2/0.55 = 4/11.

Question ITM-P03-Q19 (Infosys System Engineer ITM Mock Paper 3 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?

Options:
- **(A)** 4/33
- **(B)** 3/33
- **(C)** 5/33
- **(D)** 4/29

Sokware effort estimation has resisted four decades of methodological refinement, and the reason is not, as oken supposed, a lack of adequate techniques. Function point analysis, story point velocity tracking and Monte Carlo simulation are all, in controlled retrospective studies, reasonably well calibrated. The persistent gap between estimated and actual delivery dates arises instead from a structural asymmetry: the costs of underestimating are diffuse and delayed, distributed across a project's eventual overrun, whereas the costs of overestimating are immediate and attributable, since a padded estimate can lose a bid or provoke management scrutiny before the work even begins. Estimators therefore face an incentive gradient that has nothing to do with statistical accuracy and everything to do with organisational consequence, and this gradient reliably pushes estimates toward optimism regardless of how sophisticated the underlying model is. Attempts to correct this by adding explicit contingency buffers frequently fail for a related reason: once a buffer becomes visible in a schedule, it is treated by stakeholders as slack to be reclaimed rather than risk to be reserved, a phenomenon sometimes called the student syndrome, in reference to the tendency to begin a task only as its deadline approaches.

Correct Answer: Option **(A)** — `4/33`

Detailed Step-by-Step Solution & Mathematical Derivation:
Favourable = C(5,2)×C(4,2) = 10×6 = 60. Total ways = C(12,4) = 495. Probability = 60/495 = 4/33.

Question ITM-P03-Q20 (Infosys System Engineer ITM Mock Paper 3 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ₹40/kg and ₹60/kg) works out to ₹48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.

Options:
- **(A)** 38.9%
- **(B)** 35%
- **(C)** 41.7%
- **(D)** 33.3%

Correct Answer: Option **(A)** — `38.9%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Marked-up SP for a claimed 1 kg = 48×1.25 = ₹60. True cost of the 900 g actually given = 0.9×48 = ₹43.2. Profit = 60−43.2 = ₹16.8 on a true cost of ₹43.2, giving 16.8/43.2×100 ≈ 38.9%.

Question ITM-P03-Q21 (Infosys System Engineer ITM Mock Paper 3 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All

three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?

Options:
- **(A)** 4 hours
- **(B)** 3 hours
- **(C)** 5 hours
- **(D)** 6 hours

Correct Answer: Option **(A)** — `4 hours`

Detailed Step-by-Step Solution & Mathematical Derivation:
Combined rate (A+B−leak) = 1/12+1/15−1/20 = 5/60+4/60−3/60 = 1/10 per hour. In 4 hours, 4/10 = 2/5 of the tank fills, leaving 3/5. A+B's combined rate (no leak) = 1/12+1/15 = 3/20 per hour. Time for remaining 3/5 = (3/5)÷(3/20) = 4 hours.

Question ITM-P03-Q22 (Infosys System Engineer ITM Mock Paper 3 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?

Options:
- **(A)** 190 m
- **(B)** 200 m
- **(C)** 180 m
- **(D)** 210 m

Correct Answer: Option **(A)** — `190 m`

Detailed Step-by-Step Solution & Mathematical Derivation:
Speed ratio A:B = 1000:900 = 10:9 (A beats B by 100 m in 1000 m). Similarly B:C = 10:9. So A:C = (10/9)×(10/9) = 100:81. When A covers 1000 m, C covers (81/100)×1000 = 810 m. A beats C by 1000−810 = 190 m.

Question ITM-P03-Q23 (Infosys System Engineer ITM Mock Paper 3 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.

Options:
- **(A)** 12 kg

- **(B)** 10 kg
- **(C)** 15 kg
- **(D)** 8 kg

Correct Answer: Option **(A)** — `12 kg`

Detailed Step-by-Step Solution & Mathematical Derivation:
Copper in original alloy = 0.6×20 = 12 kg; zinc = 8 kg (unchanged after adding pure copper). For 75% copper (25% zinc): 8 = 0.25×(20+x), so 20+x = 32, x = 12 kg.

Question ITM-P03-Q24 (Infosys System Engineer ITM Mock Paper 3 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
₹8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.

Options:
- **(A)** ₹20
- **(B)** ₹40
- **(C)** ₹80
- **(D)** ₹100

Correct Answer: Option **(A)** — `₹20`

Detailed Step-by-Step Solution & Mathematical Derivation:
Half-yearly compounding: 2 periods at 5% each. Amount = 8000×1.05×1.05 = 8,820, so CI = ₹820. SI = 8000×10%×1 = ₹800. Difference = 820−800 = ₹20.

Question ITM-P03-Q25 (Infosys System Engineer ITM Mock Paper 3 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In what time will ₹8,000 amount to ₹9,261 at 5% per annum, compounded annually?

Options:
- **(A)** 2 years
- **(B)** 3 years
- **(C)** 4 years
- **(D)** 3.5 years

Correct Answer: Option **(B)** — `3 years`

Detailed Step-by-Step Solution & Mathematical Derivation:
9,261/8,000 = 1.157625 = 1.05^3 (since 1.05^2=1.1025, ×1.05=1.157625). So the time is 3 years.

Question ITM-P04-Q16 (Infosys System Engineer ITM Mock Paper 4 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?

Options:
- **(A)** 43,200
- **(B)** 86,400
- **(C)** 21,600
- **(D)** 14,400

Correct Answer: Option **(A)** — `43,200`

Detailed Step-by-Step Solution & Mathematical Derivation:
Seat 6 men in a circle: (6−1)! = 120 ways. This creates 6 gaps; choose and arrange 4 women in 4 of these 6 distinct gaps: P(6,4) = 6×5×4×3 = 360. Total = 120×360 = 43,200.

Question ITM-P04-Q17 (Infosys System Engineer ITM Mock Paper 4 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?

Options:
- **(A)** 36
- **(B)** 72
- **(C)** 18
- **(D)** 144

Correct Answer: Option **(A)** — `36`

Detailed Step-by-Step Solution & Mathematical Derivation:
ORANGE has 3 vowels (O,A,E) and 3 consonants (R,N,G), matching the 3 odd and 3 even positions exactly. Vowels in odd positions: 3! = 6. Consonants in even positions: 3! = 6. Total = 6×6 = 36.

Question ITM-P04-Q18 (Infosys System Engineer ITM Mock Paper 4 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?

Options:
- **(A)** 16/69
- **(B)** 8/69
- **(C)** 23/69
- **(D)** 25/69

Correct Answer: Option **(A)** — `16/69`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(D) = 0.25×0.05 + 0.35×0.04 + 0.40×0.02 = 0.0125+0.014+0.008 = 0.0345. P(C|D) = (0.40×0.02)/0.0345 = 0.008/0.0345 = 16/69.

Question ITM-P04-Q19 (Infosys System Engineer ITM Mock Paper 4 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?

Options:
- **(A)** 3/28
- **(B)** 1/8
- **(C)** 3/56
- **(D)** 5/28

Correct Answer: Option **(A)** — `3/28`

Detailed Step-by-Step Solution & Mathematical Derivation:
Treating the 3 girls as one block: favourable arrangements = 6!×3! = 720×6 = 4,320. Total arrangements = 8! = 40,320. Probability = 4320/40320 = 3/28.

Question ITM-P04-Q20 (Infosys System Engineer ITM Mock Paper 4 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A trader buys 80 kg of rice at ₹25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?

Options:
- **(A)** ₹26.67
- **(B)** ₹25
- **(C)** ₹28
- **(D)** ₹24

Correct Answer: Option **(A)** — `₹26.67`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total CP = 80×25 = ₹2,000; target total SP (15% profit) = ₹2,300. SP from the first 50 kg (cost ₹1,250, at 20% profit) = ₹1,500. Remaining SP needed = 2300−1500 = ₹800 for 30 kg → ₹800/30 ≈ ₹26.67/kg.

Question ITM-P04-Q21 (Infosys System Engineer ITM Mock Paper 4 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?

Options:
- **(A)** 15 days
- **(B)** 14 days
- **(C)** 16 days
- **(D)** 13 days

Correct Answer: Option **(A)** — `15 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
Rates (LCM 60): A=3/60, B=2/60, C=1/60 per day. First 4 days (all three): 4×6/60=24/60 done. Next 3 days (A+C only): 3×4/60=12/60 done — total 36/60=3/5 done, 2/5=24/60 remains. A alone at 3/60/day takes 24/60÷3/60 = 8 more days. Total = 4+3+8 = 15 days.

Question ITM-P04-Q22 (Infosys System Engineer ITM Mock Paper 4 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?

Options:
- **(A)** 10:18 AM
- **(B)** 10:12 AM
- **(C)** 10:06 AM
- **(D)** 10:24 AM

Correct Answer: Option **(A)** — `10:18 AM`

Detailed Step-by-Step Solution & Mathematical Derivation:
From 8:00 to 8:30, only A moves: 60×0.5=30 km, leaving 270 km. From 8:30, combined speed = 150 km/h; time = 270/150 = 1.8 hr = 1 hr 48 min. Meeting time = 8:30 AM + 1:48 = 10:18 AM.

Question ITM-P04-Q23 (Infosys System Engineer ITM Mock Paper 4 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?

Options:
- **(A)** 30,000
- **(B)** 28,000
- **(C)** 32,000
- **(D)** 24,000

Correct Answer: Option **(A)** — `30,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Winner's margin (60%−40%=20% of votes cast) = 4,800 → votes cast = 4800/0.20 = 24,000. Since this is 80% of registered voters: registered = 24000/0.80 = 30,000.

Question ITM-P04-Q24 (Infosys System Engineer ITM Mock Paper 4 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Find the compound interest on ₹12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).

Options:
- **(A)** ₹3,860
- **(B)** ₹3,800
- **(C)** ₹3,960
- **(D)** ₹3,720

Correct Answer: Option **(A)** — `₹3,860`

Detailed Step-by-Step Solution & Mathematical Derivation:
Year 1: 12000×1.10 = 13,200 (CI so far = 1,200). For the next half-year, SI on 13,200 at 10% for 0.5 yr = 660. Total amount = 13,860; total CI = 13860−12000 = ₹3,860.

Question ITM-P04-Q25 (Infosys System Engineer ITM Mock Paper 4 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
The compound interest on a sum for 2 years is ₹1,050, and for 3 years is ₹1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ₹1,000. Find the rate of interest per annum.

Options:
- **(A)** 10%
- **(B)** 12%
- **(C)** 8%
- **(D)** 15%

Correct Answer: Option **(A)** — `10%`

Detailed Step-by-Step Solution & Mathematical Derivation:
3rd-year interest = 1655−1050 = ₹605 — but more directly: $CI_2 - SI_2 = Pr^2 = 1050 - 1000 = 50$, and $SI_2 = 2Pr = 1000$ so Pr=500. Dividing, r = 50/500 = 0.10 = 10% (and P=₹5,000, verified: $5000 \times 1.1^3 = 6{,}655$, $CI_3 = 1{,}655$ ✓).

Question ITM-P05-Q16 (Infosys System Engineer ITM Mock Paper 5 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?

Options:
- **(A)** 121
- **(B)** 126
- **(C)** 120
- **(D)** 115

Correct Answer: Option **(A)** — `121`

Detailed Step-by-Step Solution & Mathematical Derivation:
With X fixed in and Y fixed out, the remaining pool is 5 men + 4 women = 9 people, from which 4 more members are chosen: C(9,4)=126. Subtract the invalid case where all 4 additional members are men (no woman at all): C(5,4)=5. Valid = 126−5 = 121.

Question ITM-P05-Q17 (Infosys System Engineer ITM Mock Paper 5 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?

Options:
- **(A)** 7200
- **(B)** 6000
- **(C)** 8400
- **(D)** 5040

Correct Answer: Option **(A)** — `7200`

Detailed Step-by-Step Solution & Mathematical Derivation:
Valid position-pairs with C and T exactly 3 apart: 5 pairs with C before T, plus 5 with T before C = 10 total. The remaining 6 distinct letters fill the remaining 6 positions in 6!=720 ways. Total = 10×720 = 7,200.

Question ITM-P05-Q18 (Infosys System Engineer ITM Mock Paper 5 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?

Options:
- **(A)** 1/2
- **(B)** 9/20
- **(C)** 11/20
- **(D)** 2/5

Correct Answer: Option **(A)** — `1/2`

Detailed Step-by-Step Solution & Mathematical Derivation:
By symmetry, for any two distinct numbers drawn in some order, exactly half the time the second exceeds the first. Probability = 1/2 (confirmed via direct count: 45 favourable ordered outcomes out of 90).

Question ITM-P05-Q19 (Infosys System Engineer ITM Mock Paper 5 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?

Options:
- **(A)** 391/455
- **(B)** 64/455
- **(C)** 327/455
- **(D)** 201/455

Correct Answer: Option **(A)** — `391/455`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(all 4 different suits) = (4×4×4×4)/C(16,4) = 256/1820 = 64/455. P(at least 2 same suit) = 1−64/455 = 391/455.

Question ITM-P05-Q20 (Infosys System Engineer ITM Mock Paper 5 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ₹200 (costing him ₹120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?

Options:
- **(A)** 25%
- **(B)** 20%
- **(C)** 30%
- **(D)** 15%

Correct Answer: Option **(A)** — `25%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Cost for 4 items = 4×120 = ₹480. Revenue (customer pays for 3 at ₹200) = ₹600. Profit = ₹120 on a cost of ₹480, giving 120/480×100 = 25%.

Question ITM-P05-Q21 (Infosys System Engineer ITM Mock Paper 5 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ₹9,000 is distributed in proportion to each person's work contribution. How much should C be paid?

Options:
- **(A)** ₹2,000
- **(B)** ₹1,800
- **(C)** ₹2,200
- **(D)** ₹2,500

Correct Answer: Option **(A)** — `₹2,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Rates in the ratio 1/15:1/20:1/30 = 4:3:2 (over LCM 60). C's share = (2/9)×9000 = ₹2,000.

Question ITM-P05-Q22 (Infosys System Engineer ITM Mock Paper 5 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two runners start from the same point on a 600 m circular track at the same time, in opposite

directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?

Options:
- **(A)** 300 s
- **(B)** 200 s
- **(C)** 150 s
- **(D)** 600 s

Correct Answer: Option **(A)** — `300 s`

Detailed Step-by-Step Solution & Mathematical Derivation:
A takes 600/8=75 s/lap; B takes 600/10=60 s/lap. Both are back at the start simultaneously at LCM(75,60) = 300 s.

Question ITM-P05-Q23 (Infosys System Engineer ITM Mock Paper 5 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?

Options:
- **(A)** 45 litres
- **(B)** 50 litres
- **(C)** 40 litres
- **(D)** 48 litres

Correct Answer: Option **(A)** — `45 litres`

Detailed Step-by-Step Solution & Mathematical Derivation:
Using the repeated-dilution formula: Final = $80\times(1-20/80)^2 = 80\times(3/4)^2 = 80\times9/16 = 45$ litres.

Question ITM-P05-Q24 (Infosys System Engineer ITM Mock Paper 5 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?

Options:
- **(A)** 15 years
- **(B)** 20 years
- **(C)** 10 years
- **(D)** 25 years

Correct Answer: Option **(A)** — `15 years`

Detailed Step-by-Step Solution & Mathematical Derivation:
$(1+r)^5=2$ (doubling). For 8×: $(1+r)^n=8=2^3=[(1+r)^5]^3=(1+r)^{15}$, so n=15 years.

Question ITM-P05-Q25 (Infosys System Engineer ITM Mock Paper 5 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
₹20,000 is borrowed at 10% per annum compound interest (annual compounding). ₹11,000 is repaid at the end of year 1. What amount must be paid at the end of year 2 to clear the entire debt?

Options:
- **(A)** ₹12,100
- **(B)** ₹12,000
- **(C)** ₹11,500
- **(D)** ₹13,200

Correct Answer: Option **(A)** — `₹12,100`

Detailed Step-by-Step Solution & Mathematical Derivation:
End of year 1: 20000×1.10=22,000; after repaying 11,000, remaining debt = 11,000. End of year 2: 11000×1.10 = ₹12,100.

Question ITM-P06-Q16 (Infosys System Engineer ITM Mock Paper 6 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_06_page_02.png

Question Text:
In how many ways can 8 people be seated around a circular table so that two particular people are NOT seated next to each other?

Options:
- **(A)** 3600
- **(B)** 5040
- **(C)** 1440
- **(D)** 4320

Correct Answer: Option **(A)** — `3600`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total circular arrangements = 7! = 5040. Arrangements with the two people adjacent = 2×6! = 1440. Not adjacent = 5040−1440 = 3600.

Question ITM-P06-Q17 (Infosys System Engineer ITM Mock Paper 6 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_06_page_02.png

Question Text:
How many 4-digit numbers (1000–9999) with all distinct digits are divisible by 5?

Options:
- **(A)** 952
- **(B)** 1008
- **(C)** 896
- **(D)** 504

Correct Answer: Option **(A)** — `952`

Detailed Step-by-Step Solution & Mathematical Derivation:
Last digit 0: 9×8×7 = 504. Last digit 5: the first digit has 8 choices (not 0 or 5), then 8 and 7 for the middle: 8×8×7 = 448. Total = 504+448 = 952.

Question ITM-P06-Q18 (Infosys System Engineer ITM Mock Paper 6 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
Three fair dice are rolled. Given that at least one die shows a 6, what is the probability that the sum of the three dice is 12?

Options:
- **(A)** 15/91
- **(B)** 25/216
- **(C)** 1/8
- **(D)** 10/91

Correct Answer: Option **(A)** — `15/91`

Detailed Step-by-Step Solution & Mathematical Derivation:
Outcomes with at least one 6: $6^3 - 5^3 = 91$. Of these, the sum is 12 in 15 outcomes (permutations of {6,5,1}: 6; {6,4,2}: 6; {6,3,3}: 3). Probability = 15/91.

Question ITM-P06-Q19 (Infosys System Engineer ITM Mock Paper 6 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A disease affects 1% of a population. A test detects the disease in 95% of those who have it, but also gives a false positive for 10% of those who do not. If a randomly chosen person tests positive, what is the probability that they actually have the disease?

Options:
- **(A)** 19/217
- **(B)** 19/100

- **(C)** 95/1000
- **(D)** 1/11

Correct Answer: Option **(A)** — `19/217`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(disease and positive) = 0.01×0.95 = 0.0095; P(no disease and positive) = 0.99×0.10 = 0.099. P(disease | positive) = 0.0095/0.1085 = 19/217 (≈ 8.8%).

Question ITM-P06-Q20 (Infosys System Engineer ITM Mock Paper 6 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A merchant sells two articles at ₹1,200 each. On one he gains 25%, on the other he loses 20%. What is his overall result (approximately)?

Options:
- **(A)** 2.4% loss
- **(B)** 2.4% profit
- **(C)** No gain, no loss
- **(D)** 5% loss

Correct Answer: Option **(A)** — `2.4% loss`

Detailed Step-by-Step Solution & Mathematical Derivation:
CP_1 = 1200/1.25 = 960; CP_2 = 1200/0.80 = 1500. Total CP = 2460, total SP = 2400, loss = 60. Loss % = 60/2460 ≈ 2.4% loss.

Question ITM-P06-Q21 (Infosys System Engineer ITM Mock Paper 6 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A is twice as efficient as B, and B is three times as efficient as C. Working together, all three complete a job in 9 days. In how many days can C alone complete it?

Options:
- **(A)** 90 days
- **(B)** 81 days
- **(C)** 60 days
- **(D)** 72 days

Correct Answer: Option **(A)** — `90 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let C = 1 unit/day, B = 3, A = 6, so together 10 units/day. In 9 days the work is 90 units, so C alone needs 90/1 = 90 days.

Question ITM-P06-Q22 (Infosys System Engineer ITM Mock Paper 6 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A 180 m long train passes a man walking at 6 km/h in the same direction in 12 seconds. How long will this train take to completely cross a 420 m long platform?

Options:
- **(A)** 36 s
- **(B)** 30 s
- **(C)** 40 s
- **(D)** 42 s

Correct Answer: Option **(A)** — `36 s`

Detailed Step-by-Step Solution & Mathematical Derivation:
Relative speed = 180/12 = 15 m/s = 54 km/h; the train's speed = 54+6 = 60 km/h = 50/3 m/s. Time to cross the platform = (180+420)/(50/3) = 36 s.

Question ITM-P06-Q23 (Infosys System Engineer ITM Mock Paper 6 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A man spends 40% of his salary on food, 25% of the remainder on rent, and 20% of what is then left on travel. He is left with ₹3,600. What is his salary?

Options:
- **(A)** ₹10,000
- **(B)** ₹12,000
- **(C)** ₹9,000
- **(D)** ₹11,000

Correct Answer: Option **(A)** — `₹10,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
After food: 60%. After rent (25% of 60% = 15%): 45%. After travel (20% of 45% = 9%): 36%. 36% of salary = 3600, so the salary = ₹10,000.

Question ITM-P06-Q24 (Infosys System Engineer ITM Mock Paper 6 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
A loan is repaid in two equal annual installments of ₹2,420 each, with interest at 10% per annum compounded annually (first installment due at the end of year 1). What was the loan amount?

Options:
- **(A)** ₹4,200
- **(B)** ₹4,400
- **(C)** ₹4,000
- **(D)** ₹4,840

Correct Answer: Option **(A)** — `₹4,200`

Detailed Step-by-Step Solution & Mathematical Derivation:
Loan = 2420/1.1 + 2420/1.1^2 = 2200 + 2000 = ₹4,200.

Question ITM-P06-Q25 (Infosys System Engineer ITM Mock Paper 6 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_06_page_03.png

Question Text:
In what ratio must milk-water mixtures of strength 3:2 and 5:1 (milk:water) be mixed to obtain a mixture of strength 2:1?

Options:
- **(A)** 5:2
- **(B)** 2:5
- **(C)** 3:2
- **(D)** 1:3

Correct Answer: Option **(A)** — `5:2`

Detailed Step-by-Step Solution & Mathematical Derivation:
Milk fractions: 3/5, 5/6, target 2/3. By alligation, the amounts are in the ratio (5/6−2/3) : (2/3−3/5) = 1/6 : 1/15 = 5:2.

Question ITM-P07-Q16 (Infosys System Engineer ITM Mock Paper 7 - Q16)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
In how many ways can the letters of the word SUCCESS be arranged so that no two S's are adjacent?

Options:
- **(A)** 120
- **(B)** 60
- **(C)** 240
- **(D)** 420

Correct Answer: Option **(A)** — `120`

Detailed Step-by-Step Solution & Mathematical Derivation:
Arrange the non-S letters U, C, C, E first: 4!/2! = 12 ways. This creates 5 gaps, and the 3 S's go into 3 different gaps: C(5,3) = 10. Total = 12×10 = 120.

Question ITM-P07-Q17 (Infosys System Engineer ITM Mock Paper 7 - Q17)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
A fair coin is tossed 6 times. What is the probability that no two heads appear consecutively?

Options:
- **(A)** 21/64
- **(B)** 13/32
- **(C)** 1/3
- **(D)** 5/16

Correct Answer: Option **(A)** — `21/64`

Detailed Step-by-Step Solution & Mathematical Derivation:
The number of binary strings of length 6 with no two consecutive heads is the Fibonacci number 21, out of $2^6 = 64$. Probability = 21/64 (confirmed by listing all 64 outcomes).

Question ITM-P07-Q18 (Infosys System Engineer ITM Mock Paper 7 - Q18)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
A shopkeeper marks his goods 40% above cost, gives a 15% discount on the marked price, and also uses a weight of 950 g in place of 1 kg. What is his actual profit percentage (approximately)?

Options:
- **(A)** 25.26%
- **(B)** 19%
- **(C)** 24%
- **(D)** 26.3%

Correct Answer: Option **(A)** — `25.26%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Take cost per kg = 100. Marked price = 140; after a 15% discount the selling price for a claimed 1 kg is 119. The customer actually gets only 950 g, which costs the shopkeeper 95. Profit = 119 − 95 = 24 on a cost of 95, so 24/95 ≈ 25.26%.

Question ITM-P07-Q19 (Infosys System Engineer ITM Mock Paper 7 - Q19)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
A and B together can do a job in 12 days, B and C together in 15 days, and C and A together in 20 days. In how many days can A alone do the job?

Options:
- **(A)** 30 days
- **(B)** 24 days
- **(C)** 20 days
- **(D)** 40 days

Correct Answer: Option **(A)** — `30 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
(A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 12/60 = 1/5, so A+B+C = 1/10. A = 1/10 − (B+C) = 1/10 − 1/15 = 1/30, so A alone needs 30 days.

Question ITM-P07-Q20 (Infosys System Engineer ITM Mock Paper 7 - Q20)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
A boat takes 6 hours to go 24 km upstream and 28 km downstream, and 6½ hours to go 30 km upstream and 21 km downstream. What is the speed of the boat in still water?

Options:
- **(A)** 10 km/h
- **(B)** 8 km/h
- **(C)** 12 km/h
- **(D)** 9 km/h

Correct Answer: Option **(A)** — `10 km/h`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let upstream speed = u and downstream speed = d. 24/u + 28/d = 6 and 30/u + 21/d = 6.5. Solving gives u = 6 and d = 14 (check: 4 + 2 = 6 ✓, 5 + 1.5 = 6.5 ✓). Speed in still water = (6+14)/2 = 10 km/h.

Question ITM-P07-Q21 (Infosys System Engineer ITM Mock Paper 7 - Q21)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
In an examination, 70% of the candidates passed in Mathematics, 60% passed in Science, and 15% failed in both subjects. What percentage of the candidates passed in both subjects?

Options:
- **(A)** 45%
- **(B)** 40%
- **(C)** 50%
- **(D)** 55%

Correct Answer: Option **(A)** — `45%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Passed in at least one subject = 100 − 15 = 85%. Both = 70 + 60 − 85 = 45%.

Question ITM-P07-Q22 (Infosys System Engineer ITM Mock Paper 7 - Q22)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
The difference between the compound interest (annual compounding) and the simple interest on a sum for 3 years at 20% per annum is ₹640. What is the sum?

Options:
- **(A)** ₹5,000
- **(B)** ₹4,000
- **(C)** ₹6,000
- **(D)** ₹5,500

Correct Answer: Option **(A)** — `₹5,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
For 3 years at 20%, CI − SI = $P(1.2^3 - 1 - 0.6) = 0.128P = 640$, so P = ₹5,000.

Question ITM-P07-Q23 (Infosys System Engineer ITM Mock Paper 7 - Q23)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
What is the remainder when 2^{2026} is divided by 7?

Options:
- **(A)** 2
- **(B)** 1
- **(C)** 4
- **(D)** 3

Correct Answer: Option **(A)** — `2`

Detailed Step-by-Step Solution & Mathematical Derivation:
$2^3 = 8 \equiv 1 \pmod 7$, so powers of 2 repeat every 3. $2026 = 3 \times 675 + 1$, so $2^{2026} \equiv 2^1 = 2$.

Question ITM-P07-Q24 (Infosys System Engineer ITM Mock Paper 7 - Q24)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_07_page_03.png

Question Text:
The present ages of A and B are in the ratio 5 : 3. After 4 years, the ratio of their ages will be 3 : 2. What is the sum of their present ages?

Options:
- **(A)** 32 years
- **(B)** 36 years
- **(C)** 40 years
- **(D)** 28 years

Correct Answer: Option **(A)** — `32 years`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let the ages be 5x and 3x. (5x+4)/(3x+4) = 3/2 gives 10x + 8 = 9x + 12, so x = 4. The ages are 20 and 12, and the sum is 32.

Question ITM-P07-Q25 (Infosys System Engineer ITM Mock Paper 7 - Q25)

Section: Section 2: Technical Ability (Quantitative) | **Page:** 4
Visual Reference Plate: paper_07_page_04.png

Question Text:
Three bells ring at intervals of 12, 15 and 20 minutes. They all ring together at 9:00 AM. How many times in all (including 9:00 AM) will they ring together between 9:00 AM and 3:00 PM, both times included?

Options:
- **(A)** 7
- **(B)** 6
- **(C)** 8
- **(D)** 5

Correct Answer: Option **(A)** — `7`

Detailed Step-by-Step Solution & Mathematical Derivation:
LCM(12, 15, 20) = 60 minutes. In 6 hours (360 minutes) the bells ring together at 0, 60, 120, 180, 240, 300 and 360 minutes: 7 times.
